Workspace Trajectory Tracking Control of Flexible Joint Robots Based on Backstepping Method

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***Abstract*—This paper addresses the issue of end-effector trajectory tracking control of multi-link flexible joint robot manipulators. A submanifold is designed to describe the ideal trajectory tracking performance. Dynamics of the robot is rewritten in terms of the manifold. Control schemes are derived using backstepping control design approach. Stability of the system is analyzed based on Lyapunov stability theory. Simulations are carried out to on a 2-link flexible robot manipulator. The results demonstrate effectiveness and usefulness of the proposed control method.**

Keywords—flexible joint robot; backstepping control method, workspace trajectory tracking;

I. INTRODUCTION

Flexible joint robot manipulators, as the ideal ones in which elasticity of the transmission devices on the joints have taken into consideration, have attracted a lot of attention of researchers. The reason lies not only in that it is related to the development of more precise industrial robots, but also in that it raises many interesting issues in control design.

Literature shows that huge variety of researchers have been worked in control of the flexible joint robot and wide variety of papers have been published during the last three decades[1] ~[6]. Recent years, backstepping control method for the robot interests many researchers. A method using backstepping approach to train a neural network is proposed for control of a flexible joint robot [7]. A backstepping method for tracking control of multi-joint variable stiffness robots is presented [8]. An output feedback control based on the backstepping approach is proposed. The control method guarantees asymptotic stability of the flexible joint robot system [9].

In this paper, a new workspace trajectory control method is proposed for multi-link flexible joint robots. Based on this method, the control design can be broken down into three stages stated as follows. First, a submanifold is designed to prescribe ideal end-effector trajectory control performance. Second, the dynamic model of the robot is reformatted according to the submanifold. Finally, three control laws are derived using the backstepping control method for trajectory control of the end-effector and vibration suppressing of the flexible joints. Stability of the system is analyzed based on Lyapunov stability theory. Simulations are carried on a two-link flexible robot as the testbed. The results demonstrate the effectiveness and usefulness of the proposed control method.

This paper is organized as follows. Kinematic and dynamic models are given in Section II. Section III describes the details of the control design. Stability analysis is presented in Section IV. Section V demonstrates the simulation results. Finally, conclusions are stated in Section VI.

II. KINEMATICS AND DYNAMICS OF FLEXIBLE ROBOTS

Consider an n-link flexible joint robot manipulator, each joint being modeled as a liner rotation spring between the actuator and the link that is driven by the joint. The mass and moment of inertia of the transmission are equivalently added to both sides of the actuator and link. Fig.1 shows the physics model of the *i*th joint.

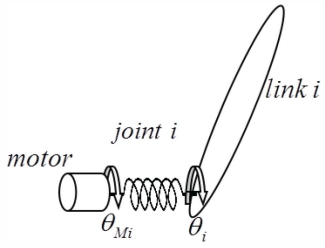


Fig. 1. Model of flexible joint.

The end-effector position of the robot is given as

$$\mathbf{P} = \mathbf{f}(\boldsymbol{\theta}) \quad \in R^N \tag{1}$$

where $\boldsymbol{\theta} \in R^n$ is the joint variable of the link side with θ_i being its *i*th element. Taking time derivative of (1) one obtains end-effector velocity as

$$\dot{\mathbf{P}} = \mathbf{J}(\boldsymbol{\theta})\dot{\boldsymbol{\theta}} \tag{2}$$

where $\mathbf{J}(\boldsymbol{\theta}) \in R^{N\times n}$ is Jacobian. Similarly, we have end-effector acceleration as follows.

$$\ddot{\mathbf{P}} = \mathbf{J}(\boldsymbol{\theta})\ddot{\boldsymbol{\theta}} + \dot{\mathbf{J}}(\boldsymbol{\theta})\dot{\boldsymbol{\theta}} \tag{3}$$

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Using the Lagrange formulation, dynamics of the robot is derived as follows.

$$\mathbf{M}_L(\boldsymbol{\theta})\ddot{\boldsymbol{\theta}} + \mathbf{D}_L(\boldsymbol{\theta},\dot{\boldsymbol{\theta}})\dot{\boldsymbol{\theta}} - \mathbf{K}(\boldsymbol{\theta}_M - \boldsymbol{\theta}) + \mathbf{g}_L(\boldsymbol{\theta}) = \mathbf{0} \tag{4}$$

$$\mathbf{I}_M\ddot{\boldsymbol{\theta}}_M + \mathbf{K}(\boldsymbol{\theta}_M - \boldsymbol{\theta}) = \mathbf{u} \tag{5}$$

where, (4) and (5) are motion equations with respect to the link side and actuator side, $\boldsymbol{\theta} \in R^n$ and $\boldsymbol{\theta}_M \in R^n$ are the joint variables with respect to both sides, $\mathbf{u} \in R^n$ is the actuator torque applied at the joints, $\mathbf{M}_L(\boldsymbol{\theta}) \in R^{n\times n}$ and $\mathbf{D}_L(\boldsymbol{\theta},\dot{\boldsymbol{\theta}})\dot{\boldsymbol{\theta}} \in R^n$ are inertia matrix and Coriolis and centrifugal forces, $\mathbf{I}_M \in R^{n\times n}$ and $\mathbf{K} \in R^{n\times n}$ are the actuator side inertia matrix and the joint stiffness matrix, $\mathbf{g}_L(\boldsymbol{\theta}) \in R^n$ is the gravity. It should be pointed out that $\mathbf{I}_M$ and $\mathbf{K}$ are diagonal matrices with the diagonal elements being the moment of inertia and spring constant of each joint, respectively.

Note that $\mathbf{M}_L(\boldsymbol{\theta})$ is a positive definite symmetric matrix and $\frac{1}{2}\dot{\mathbf{M}}_L(\boldsymbol{\theta}) - \mathbf{D}_L(\boldsymbol{\theta},\dot{\boldsymbol{\theta}})$ is skew symmetric matrix.

III. Control System Design

The main difference between the flexible joint robot and its ordinary counterpart is that links of the former is not directly driven by the actuators but through the joint flexible behavior. Hence, the control design should focus on two issues: one is how to cause such the joint flexible behavior that it meets the requirements of desired link motion and further end-effector motion, another is how to dampen vibration of the flexible link to ensure stability of the system. The control objective in this research is to achieve precise manipulation tasks of the flexible joint robot. In details, the control system has to play the role of ensuring precise end-effector trajectory tracking while keeping the whole system stable.

In this section, we design a manifold to prescribe the ideal trajectory tracking performance, and then we derive the control strategy based on backstepping control approach such that motion of the robot converges to the manifold while keeping the system stable.

A. Submanifold Design

Let $\mathbf{P}_d$, $\dot{\mathbf{P}}_d$, $\ddot{\mathbf{P}}_d$ be the planned end-effector trajectories required by the manipulation task. The tracking errors with respect to position and velocity are given as $\mathbf{e} = \mathbf{P} - \mathbf{P}_d$ and $\dot{\mathbf{e}} = \dot{\mathbf{P}} - \dot{\mathbf{P}}_d$. The submanifold is design as

$$\mathbf{S} = \dot{\mathbf{e}} + \boldsymbol{\lambda}\mathbf{e} = \mathbf{0} \tag{6}$$

where $\boldsymbol{\lambda} \in R^{N\times N}$ is a positive-definite matrix and can be chosen as a diagonal matrix. Obviously, precise trajectory tracking is achieved when the motion of the robot converges and remains on the manifold.

We rewrite the motion equation (4) in terms of the submanifold and let (5) remain unchanged. The new formulation of the dynamics is given as below.

$$\begin{aligned}&\mathbf{M}_P(\boldsymbol{\theta})\dot{\mathbf{S}} + \mathbf{D}_P(\boldsymbol{\theta},\dot{\boldsymbol{\theta}})\mathbf{S} + \mathbf{h}_P(\boldsymbol{\theta},\dot{\boldsymbol{\theta}}) + \mathbf{M}_P(\boldsymbol{\theta})(\ddot{\mathbf{P}}_d - \boldsymbol{\lambda}\dot{\mathbf{e}}) \\ &+ \mathbf{D}_P(\boldsymbol{\theta},\dot{\boldsymbol{\theta}})(\dot{\mathbf{P}}_d - \boldsymbol{\lambda}\mathbf{e}) - \mathbf{J}^{-T}(\boldsymbol{\theta})\mathbf{K}(\boldsymbol{\theta}_M - \boldsymbol{\theta}) = \mathbf{0}\end{aligned} \tag{7}$$

$$\mathbf{I}_M\ddot{\boldsymbol{\theta}}_M + \mathbf{K}(\boldsymbol{\theta}_M - \boldsymbol{\theta}) = \mathbf{u} \tag{8}$$

$\mathbf{M}_P(\boldsymbol{\theta})$ and $\mathbf{D}_P(\boldsymbol{\theta})$ are generalized inertia matrix and Coriolis and centrifugal coincident matrix described in the work space. In detail, they are defined as below.

$$\mathbf{M}_P(\boldsymbol{\theta}) = \mathbf{J}^{-T}(\boldsymbol{\theta})\mathbf{M}(\boldsymbol{\theta})\mathbf{J}^{-1}(\boldsymbol{\theta}) \tag{9}$$

$$\mathbf{D}_P(\boldsymbol{\theta}) = \mathbf{J}^{-T}(\boldsymbol{\theta})\mathbf{D}(\boldsymbol{\theta})\mathbf{J}^{-1}(\boldsymbol{\theta}) \tag{10}$$

where $\mathbf{J}^{-1}(\boldsymbol{\theta})$ and $\mathbf{J}^{-T}(\boldsymbol{\theta})$ denote invert and invert transpose matrices of the Jacobian.

In (7), $\mathbf{h}_P(\boldsymbol{\theta},\dot{\boldsymbol{\theta}})$ is combined with gravity and nonlinear forces described in the workspace. The detail is given as follows.

$$\mathbf{h}_P(\boldsymbol{\theta},\dot{\boldsymbol{\theta}}) = \mathbf{J}^{-T}(\boldsymbol{\theta})(\mathbf{g}(\boldsymbol{\theta}) - \mathbf{M}(\boldsymbol{\theta})\mathbf{J}^{-1}(\boldsymbol{\theta})\dot{\mathbf{J}}(\boldsymbol{\theta})\dot{\boldsymbol{\theta}}) \tag{11}$$

The second term on the right of above equation is nonlinear force caused by the coordinate transformation from the joint space to workspace.

Note that since $\frac{1}{2}\dot{\mathbf{M}}_L(\boldsymbol{\theta}) - \mathbf{D}_L(\boldsymbol{\theta},\dot{\boldsymbol{\theta}})$ is a skew symmetric matrix, it is evident that $\frac{1}{2}\dot{\mathbf{M}}_P(\boldsymbol{\theta}) - \mathbf{D}_P(\boldsymbol{\theta},\dot{\boldsymbol{\theta}})$ is also a skew symmetric matrix.

B. Design of the First Control Law

From (7) it is seen that the links of the robot are essentially driven by the elastic torque, say flexible behavior, of joints. Thus we can treat the elastic torque as nominal control input and express it as

$$\boldsymbol{\tau} = \mathbf{K}(\boldsymbol{\theta}_M - \boldsymbol{\theta}) \tag{12}$$

then, design the control law accordingly. The design of the first control law is aimed at end-effector trajectory tracking. The details are given as follows.

$$\begin{aligned}\boldsymbol{\tau}_d = \mathbf{J}^T(\boldsymbol{\theta})(&\mathbf{M}_P(\boldsymbol{\theta})(\ddot{\mathbf{P}}_d - \boldsymbol{\lambda}\dot{\mathbf{e}}) + \mathbf{D}_P(\boldsymbol{\theta},\dot{\boldsymbol{\theta}})(\dot{\mathbf{P}}_d - \boldsymbol{\lambda}\mathbf{e}) \\ &+ \mathbf{h}_P(\boldsymbol{\theta},\dot{\boldsymbol{\theta}}) - \mathbf{K}_P\mathbf{S})\end{aligned} \tag{13}$$

where $\mathbf{K}_P \in R^{N\times N}$ is a feedback gain matrix, to be designed as a diagonal positive-definite matrix.

(13) presents a resolved-acceleration-control-like control scheme with adding a linear feedback term of the manifold to it.

Under the first control law, (7) becomes

$$\mathbf{M}_P(\boldsymbol{\theta})\dot{\mathbf{S}} + \mathbf{D}_P(\boldsymbol{\theta},\dot{\boldsymbol{\theta}})\mathbf{S} + \mathbf{K}_P\mathbf{S} = \mathbf{J}^{-T}(\boldsymbol{\theta})\mathbf{e}_\tau \tag{14}$$

where $\mathbf{e}_\tau = \boldsymbol{\tau} - \boldsymbol{\tau}_d$ denotes the error between the real elastic torque and the nominal control input.

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C. Design of the Second Control Law

The objective of the second control law design is to ensure the error of the first control input $\mathbf{e}_\tau$ to converge to zero. Taking derivative of (12) with respect to time one obtains

$$\dot{\boldsymbol{\tau}} = \mathbf{K}(\dot{\boldsymbol{\theta}}_M - \dot{\boldsymbol{\theta}}) \tag{15}$$

The second control law is designed to prescribe the desired joint motion on the motor side so that the abovementioned objective is realized. The details are given as follows.

$$\dot{\boldsymbol{\theta}}_{Md} = \mathbf{K}^{-1}(\dot{\boldsymbol{\tau}}_d - \mathbf{K}_t\mathbf{e}_\tau - \mathbf{J}^{-1}(\boldsymbol{\theta})\mathbf{S}) + \dot{\boldsymbol{\theta}} \tag{16}$$

Under this control law (15) becomes

$$\dot{\mathbf{e}}_\tau + \mathbf{K}_t\mathbf{e}_\tau = \mathbf{K}\mathbf{e}_\theta - \mathbf{J}^{-1}(\boldsymbol{\theta})\mathbf{S} \tag{17}$$

where $\mathbf{e}_\theta = \dot{\boldsymbol{\theta}}_M - \dot{\boldsymbol{\theta}}_{Md}$ denotes the error between the real motion and the desired motion of the joint on the actuate side.

D. Design of the Third Control Law

The third control law is to guarantee that the real joint motion converges to the ideal joint motion.

$$\mathbf{u} = \mathbf{I}_m\ddot{\boldsymbol{\theta}}_{Md} - \mathbf{K}_\theta\mathbf{e}_\theta + \boldsymbol{\tau}_d \tag{18}$$

where $\mathbf{K}_\theta \in R^{n\times n}$ is positive-definite feedback gain matrix. Under the control law, subsystem (8) becomes

$$\mathbf{I}_m\dot{\mathbf{e}}_\theta + \mathbf{K}_\theta\mathbf{e}_\theta + \mathbf{e}_\tau = \mathbf{0} \tag{19}$$

It should be pointed out that the control input generated by the third control law is the actual control input to the robot, and it combined the former two control laws together.

IV. Stability Analysis

Stability of the whole system is analyzed based on Lyapunov stability theory. In so doing, we first define Lyapunov function candidate as follows.

$$V = \frac{1}{2}\mathbf{S}^T\mathbf{M}_P(\boldsymbol{\theta})\mathbf{S} + \frac{1}{2}\mathbf{e}_\tau^T\mathbf{e}_\tau + \frac{1}{2}\mathbf{e}_\theta^T\mathbf{K}\mathbf{I}_m\mathbf{e}_\theta > 0 \tag{20}$$

Taking derivative with respect to time one obtains,

$$\dot{V} = \mathbf{S}^T\mathbf{M}_P(\boldsymbol{\theta})\dot{\mathbf{S}} + \frac{1}{2}\mathbf{S}^T\dot{\mathbf{M}}_P(\boldsymbol{\theta})\mathbf{S} + \mathbf{e}_\tau^T\dot{\mathbf{e}}_\tau + \mathbf{e}_\theta^T\mathbf{K}\mathbf{I}_m\dot{\mathbf{e}}_\theta \tag{21}$$

Substituting solution trajectories of (14), (17) and (19) into (21) yields,

$$\begin{aligned}\dot{V} = {} & \mathbf{S}^T(-\mathbf{D}_P(\boldsymbol{\theta},\dot{\boldsymbol{\theta}})\mathbf{S} - \mathbf{K}_P\mathbf{S} + \mathbf{J}^{-T}(\boldsymbol{\theta})\mathbf{e}_\tau) + \frac{1}{2}\mathbf{S}^T\dot{\mathbf{M}}_P(\boldsymbol{\theta})\mathbf{S} \\ & + \mathbf{e}_\tau^T(-\mathbf{K}_t\mathbf{e}_\tau + \mathbf{K}\mathbf{e}_\theta - \mathbf{J}^{-1}(\boldsymbol{\theta})\mathbf{S}) \\ & + \mathbf{e}_\theta^T\mathbf{K}(-\mathbf{K}_\theta\mathbf{e}_\theta - \mathbf{e}_\tau)\end{aligned} \tag{22}$$

Noting that since $\frac{1}{2}\dot{\mathbf{M}}_P(\boldsymbol{\theta}) - \mathbf{D}_P(\boldsymbol{\theta},\dot{\boldsymbol{\theta}})$ is a skew symmetric matrix, one has

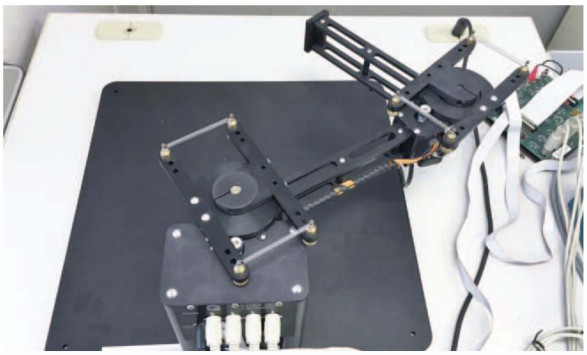


Fig. 2. 2-link flexible joint robot

$$\dot{V} = -\mathbf{S}^T\mathbf{K}_P\mathbf{S} - \mathbf{e}_\tau^T\mathbf{K}_t\mathbf{e}_\tau - \mathbf{e}_\theta^T\mathbf{K}\mathbf{K}_\theta\mathbf{e}_\theta < 0 \tag{23}$$

Thus, stability of the whole system is proved.

V. Trajectory Control Simulations

In order to demonstrate the effectiveness of the proposed control method end-effector trajectory tracking control simulations are carried out on a 2-link flexible joint robot arm made by Quanser Corporation shown in Fig.2. The parameters of the robot dynamics have been identified based on identification experiments.

The control strategy shown in section III is used to establish the control system. The end-effector trajectories are planned as a straight line in the workspace with velocity trajectories on X and Y directions being planned as trigonometric functions and zero initial value and zero final value of velocities in both directions are preset. Simulations are carried out on Matlab. The sampling time set as 1ms, and control time set as 4s. Fig.3 ~ Fig.7 show a trajectory tracking control simulation result.

From the simulation results, it is seen that precise end-effector trajectory tracking is achieved and joint vibration is effectively damped. It demonstrates that under the proposed control method, high performance of the control system can be guaranteed.

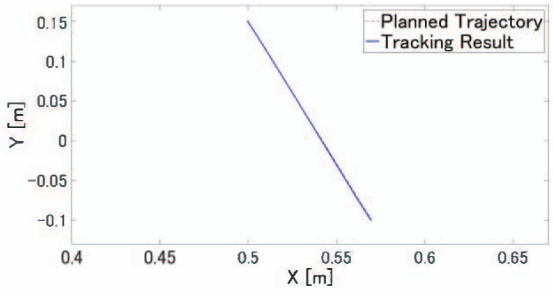


Fig. 3. Planned trajectory and tracking result in the workspace.

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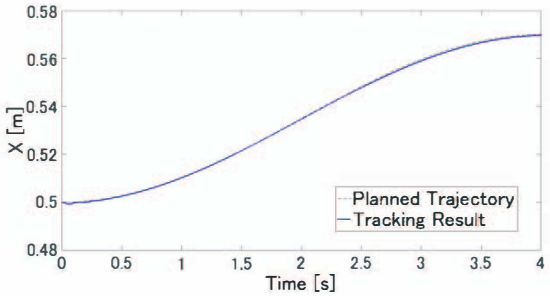


Fig. 4. Time history of planned trajectory and tracking result in X direction.

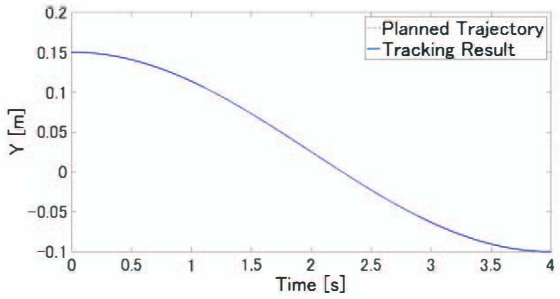


Fig. 5. Time history of planned trajectory and tracking result in Y direction.

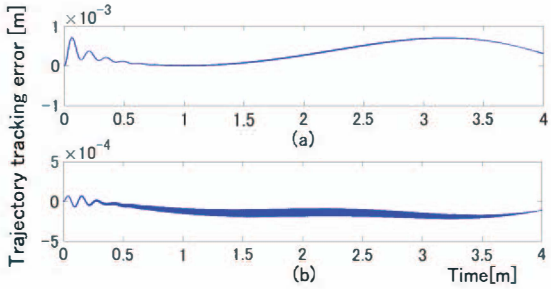


Fig. 6. Time history of tracking error, (a) and (b): X and Y directions.

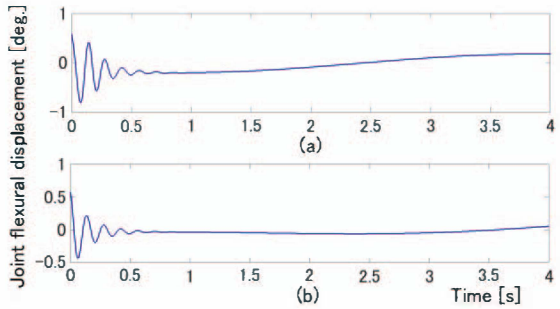


Fig. 6. Time history of joint vibration, (a): joint1, (b): joint2 .

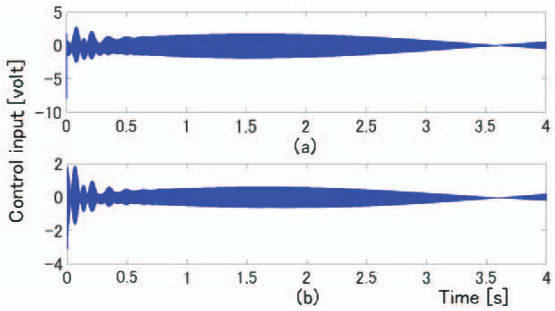


Fig. 7. Time history of control inputs, (a): joint1, (b): joint2 .

VI. Conclusions

This paper presented a new control method for workspace trajectory tracking control of multi-link flexible joint robots. An ideal submanifold was designed to prescibe the trajectory tracking performance. Control laws were derived based on backstepping control approach. Stability of the control system was verified using Lyapunov stability theory. Simulations were carried out on a 2-link flexible joint robot. Since the time derivitaves in the control laws were calculated by using the simple numerical differenatiation method, the control input signals contained high frequency noises. Neverthless, the simulation results demonstrated high performance of the system in end-effector trajectory tracking and joint vibration damping.

References

[1] Z. H. Jiang, S. Higaki: "Control of flexible joint robot manipulators using a combined controller with neural network and linear regulator," J. of Systems and Control Engineering, vol.255, part1, pp.798-806, 2011

[2] C. Liu, C. C. Cheah and J. J. E. Slotine, "Adaptive task-space regulation of rigid-link flexible-joint robots with uncertain kinematics," Automatica, vol.44, pp.1806-1814, 2008.

[3] M. W. Spong, "Modeling and control of elastic joint robots," J. Dynamic Systems Measurement and Control, vol. 109, no. 4, pp. 310–319, 1987.

[4] S. S. Ge, "Adaptive control design for flexible joint manipulators," *Automatica*, vol. 32, no. 2, pp. 273-278, 1996.

[5] A. De Luca, B. Siciliano, and L. Zollo, "PD control with on-line gravity compensation for robots with elastic joints: Theory and experiments," Automatica, vol. 41, no. 10, pp. 1809–1819, 2005.

[6] N. Yun and J-B Su, "Design of a Disturbance Observer for a Two-Link Manipulator with Flexible Joints", IEEE Trans. on Control Systems Technology, vol.22, no.2, pp.809-815, 2014.

[7] C.J.B. Macnab, G.M.T. D'Elouterio, and M. Meng, "CMAC Adaptive Control of Flexible-Joint Robots Using Backstepping with Tuning Functions," Proc. IEEE Int. Conf. Rob. Autom., Vol. 3, New Orleans, pp. 2679-2686, 2004.

[8] F. Petit, A. Daasch, and A. Albu-Schäffer: "Backstepping Control of Variable Stiffness Robots," IEEE Trans. on Control Systems Technology, vol.23, No.6, pp.2195-2202, 2015.

[9] Antonio Loria, S. Avila-Becerril: "Output-feedback global tracking control of robot manipulators with flexible joints," Proc. Of American Control Conference, pp.4032-4037, 2014.

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